

PRESENTATION LAYER

Chat Interface

Interactive dialogue
tutoring interface

Streamlit

Mode Selector

Tutor / Hints / Exercise
generation modes

Progress Dashboard

Knowledge mastery
visualization

Session Manager

Chat history and
student profiles

REST API / JSON



APPLICATION LAYER

Dialogue Tutor

Socratic guided
instruction with
context-aware
responses

Prompt Engineering

Adaptive Hints

Four-level progressive
hint hierarchy
with automatic
escalation
after 3 consecutive
failures

L1 → L2 → L3 → L4

Exercise Generator

Topic and difficulty-
based
problem generation

JSON Output

Knowledge Tracker

Bayesian mastery
update with
weak point
identification

$$M = \frac{C + P}{T + W}$$

Prompt Construction + Context Injection



INFERENCE LAYER

Prompt Engine

System prompts with chat
history and
knowledge context

Qwen2.5-7B-Instruct

FP16 inference with automatic
device
mapping via Transformers

RTX 3090 · 24GB VRAM

Tokenizer

Chat template application and
token
encoding/decoding

SQLAlchemy ORM (Object-Relational Mapping)



DATA LAYER

Chat History

Messages, roles, and
timestamps

Knowledge State

Mastery levels per
knowledge point

Exercise Records

Performance metrics
and attempt
history

Session Data

Student profiles and
preferences

SQLite